Dr. Joshua L. Pughe-Sanford

youtube | email | website | scholar

Theoretical, computational, and mathematical physicist specializing in nonlinear dynamics and theories of learning and inference in neural systems

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Experience

Research Scientist 2026–Present
Principia Princeton, NJ

Flatiron Research Fellow 2023–2026
Flatiron Institute, Center for Computational Neuroscience New York, NY

Education

Ph.D. in Physics · M.S. in Mathematics (GPA: 3.9) 2017–2023
Georgia Institute of Technology, Center for Non-Linear Dynamics Atlanta, GA

B.S. in Physics · Minor in Mathematics (GPA: 3.9, Highest Honors) 2013–2017
Emory University Atlanta, GA

Awards

Presidential Fellow · Herbert P. Haley Fellow · Phi Beta Kappa Honors Society

Teaching and Mentorship Experience

Flatiron Institute *2023–2026*

Mentored six graduate students, resulting in publication [1, 4].

Georgia Institute of Technology *2017–2023*

Mentored three undergraduate students, resulting in publication [6].

Promoted to lead teaching assistant, training and managing 30 graduate TAs.

Received on average **4.9/5** on teaching evaluations.

Taught graduate lectures on nonlinear dynamics and mathematical methods for physicists.

Service

Diversity, Equity, and Inclusion Committee, *at Simons Foundation* *2024–2026*

Advocated for more inclusive hiring practices across the institution.

Led intern selection for my research group from 500+ applicants, prioritizing candidates underrepresented in science.

Physics Allies for Wellness, *at Georgia Institute of Technology* *2020–2023*

Co-founded an organization providing support, resources, and action plans for those facing injustice and inequity in the physics department.

Graduate Association of Physicists, *at Georgia Institute of Technology* *2018–2020*

Publications † indicates co-first authorship

[1] I. Kalburge, **J. L. Pughe-Sanford**, and A. Williams, "Sequential regression in Recurrent Neural Networks," *submitted*.

[2] **J. L. Pughe-Sanford**†, A. Mah†, S. Harvey, and A. Williams, "Linear approximations to HMM filtering," *NeurIPS* (2026).

[3] F. Noah, **J. L. Pughe-Sanford**, and S. Grauer, "Degree-of-freedom and optimization dynamic effects on the observability of the Kuramoto-Sivashinsky model," *JCP* (2026).

[4] S. Qin, **J. L. Pughe-Sanford**, A. Genkin, P. G. Ozdil, P. Greengard, A. M. Sengupta, and D. Chklovskii, "A network of biologically inspired Rectified Spectral Units (ReSUs) learns hierarchical features without error backpropagation," *AAAI*, 2026.

[5] **J. L. Pughe-Sanford**†, X. Ding†, J. J. Moore†, C. Epstein, P. Greengard, A. M. Sengupta, and D. Chklovskii, "Neurons as Detectors of Coherent Sets in Sensory Dynamics," *NeurIPS*, 2025.

[6] **J. L. Pughe-Sanford**, S. Quinn, T. Balabanski, and R. O. Grigoriev, "Computing chaotic time-averages from few periodic or non-periodic orbits," *Chaos*, 2025.

[7] J. J. Moore, A. Genkin, M. Tournoy, **J. L. Pughe-Sanford**, R. R. de Ruyter van Steveninck, and D. B. Chklovskii, "The Neuron as a Direct Data-Driven Controller," *PNAS*, 2024.

[8] **J. L. Pughe-Sanford**, "On the Generalization of Shadowing to Fluid Turbulence: Practical Methods For Quantifying Dynamical Similarity," Ph.D. Dissertation, Georgia Institute of Technology, 2023.

[9] **J. L. Pughe-Sanford**†, C. J. Crowley†, W. Toler, R. O. Grigoriev, and M. F. Schatz, "Observing a dynamical skeleton of turbulence in Taylor–Couette flow experiments," *Philosophical Transactions of the Royal Society A*, 2023.

[10] **J. L. Pughe-Sanford**†, C. J. Crowley†, W. Toler, M. C. Krygier, R. O. Grigoriev, and M. F. Schatz, "Turbulence tracks recurrent solutions," *PNAS*, 2022.

[11] M. C. Krygier, **J. L. Pughe-Sanford**, and R. O. Grigoriev, "Exact coherent structures and shadowing in turbulent Taylor–Couette flow," *Journal of Fluid Mechanics*, 2021.

[12] S. Boettcher and **J. L. Pughe-Sanford**, "Renormalization of Discrete-Time Quantum Walks with non-Grover Coins," *Journal of Statistical Mechanics: Theory and Experiment*, 2017.

[13] **J. L. Pughe-Sanford**, "Properties of Quantum Walks within Various One-Dimensional Media," Honors Thesis, Emory University, 2017.

Recorded Talks

J. L. Pughe-Sanford, S. Quinn, T. Balabanski, and R. O. Grigoriev, "Computing Chaotic Time-Averages from a Small Number of Periodic Orbits." APS DFD, 2023

J. L. Pughe-Sanford, and R. O. Grigoriev, "Vortex Interactions: a Low-Dimensional Approach to the Inverse Cascade." APS DFD, 2022

J. L. Pughe-Sanford, and R. O. Grigoriev, "Heteroclinic Connections as Predictors of Extreme Events in Weakly Turbulent Flow." APS DFD, 2019

Research Summary

Principia, *with A. Saxe and J. Lee* *2026–Present*

Building theoretical model organisms of learning: architectures simple enough to study from first principles, yet sophisticated enough to explore phenomena seen in large models. Personas and emergent misalignment are of particular interest.

Flatiron Institute, *with A. Williams* *2025–2026*

Investigated how sufficient statistics are represented in recurrent neural architectures during sequential inference tasks [1].

Characterized which hidden Markov processes are easily filtered by linear RNNs, shedding light on where complicated architecture is needed and, importantly, where it is *not* [2].

Flatiron Institute, *with D. Chklovskii* *2023–2025*

Engineered interpretable, dynamical models of neural computation through the lens of observability [5] and controllability [7].

Georgia Institute of Technology, *with R. Grigoriev* *2017–2023*

Generalized the Axiom A framework of low-dimensional chaos to fluid turbulence [8]. In particular, we presented evidence that turbulence is well predicted over short time intervals by invariant sets [9–11]. We then showed that, over long time scales, ergodic averages can be computed accurate from even a meager amount of such sets [6].

Emory University, *with S. Boettcher* *2015–2017*

Derived scaling laws for the rate at which a quantum wavefunction will spread out over a self-similar lattice, and validated these results *in silico* [12].

Explored computational complexity advantages of quantum versus classical algorithms [13].

Princeton University, *with F. Calaprice* *2013–2017*

Helped design, fabricate, and deploy a dark matter detection experiment.

Automated and improved the purification process that removes radioactive contaminants from the detector, improving its sensitivity.

Oxford College, *with R. Conceicao* *2013–2015*

Analyzed the Collatz conjecture and its polynomial variant, producing asymptotic approximations for total stopping time.

Hobbies

Robotics · Video Game Development · Volleyball · Camping